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ANOMALOUS PLIOCENE PALEOMAGNETIC POLE POSITIONS FROM BAJA CALIFORNIA

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Paleomagnetic data from some fine-grained intertidal muds of Pliocene age from Baja California yield pole positions which depart significantly from other Upper Tertiary pole positions for North America, whereas similar material of Recent age yields a pole position close to that of the present magnetic pole. The sense of the displacement of the Pliocene pole position suggests that the collection site may have undergone a clockwise rotation of about 30° since the rocks were magnetized. If this is the case it suggests that portions of western North America have rotated as independent blocks in addition to the lateral translations typical of sea-floor spreading models.

1. Introduction

In a study of modern red sediments and the way in which they become magnetized, it has been possible to determine a Pliocene paleomagnetic pole position from Baja California, Mexico. As part of the study, we have investigated the remanent magnetization of a series of intertidal muds of various ages. Walker [1] had made a detailed study of the geology of these deposits. He has examined Recent material, Pleistocene material, and Pliocene material; the last of these being dated by fossil control. The fossil Ostrea California is found imbedded in the sequence but it should be noted that this is not a very diagnostic fossil. The older sediments are redder, indicating that post-depositional changes are taking place in the iron minerals. Detailed studies of the magnetic properties however do not reveal any significant difference in magnetic properties with time [2]. This leads us to believe that the material acquired its magnetization at the time that these fine mudstones were deposited and that the original magnetization has probably been retained. The remanence is carried by magnetite. Typical grains are about 1μ in size and probably grade down to sizes much less than 0.1μ, the limit of microscope observation.

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2. Magnetic measurements

The natural remanent magnetization (NRM) of

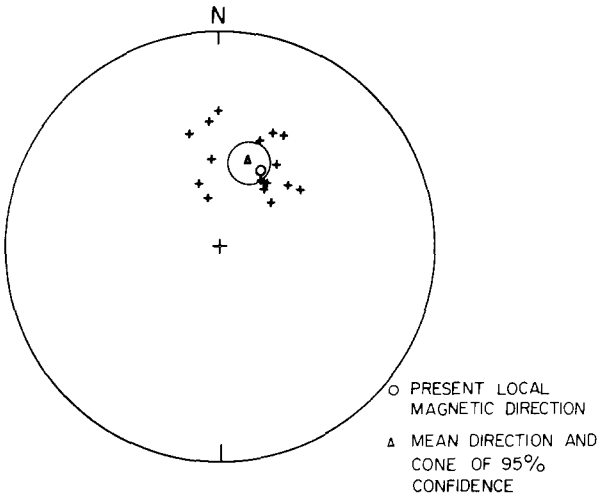


Fig. 1. Direction of magnetization in Recent muds after A.C. cleaning.

the samples was found to be scattered, indicating that the samples contain unstable magnetic components. However, with A.C. demagnetization good groupings of the directions of magnetization were achieved. Sixteen samples of Recent surface mud were collected from the basin known as Laguna Salada and from intertidal muds along the west coast of the Gulf of California (see Walker [1] and Strangway et al. [2] for precise locations). The directions of magnetization after cleaning in fields of 500 Oe have a fairly close grouping around the present magnetic field direction for this area (fig. 1). At the 95% level there is no significant difference between the direction of magnetization recorded in these Recent muds and the direction of the local magnetic field (fig. 1). This is strong support for the interpretation that the intertidal muds acquire a detrital remanent magnetization (DRM) parallel to the present magnetic field. The DRM does not show the inclination error that has been reported in the study of many artificial sediments [3].

Identical studies were done on a series of distinctly reddened muds believed to be Pliocene equivalents of the Recent muds just discussed. Fossil control indicates a Pliocene age for the muds. Two sets of samples were taken. One of these was from 35 miles south of San Felipe, Baja California on the Gulf of California at a site described by Walker [1]. In this set, three groups of samples referred to in table 1 as WRBA, WRBB, and WRBC were collected. Slight folding is present and small structural corrections up to a maximum of 12° were applied in various directions. A total of 47 samples were taken from these sites and the magnetizations after demagnetizing at 150 Oe are shown in fig. 2a, 2b and 2c and summarized in table 1. The second set of samples was taken from a locality known as Canon Rojo on the west side of the Cocopah Mountains. Here, coarse-grained redbeds predominate, but interbedded fine-grained sands and muds are found, presumably representing local, quiet-water deposition conditions. The data from these are shown in fig. 2d and also summarized in table 1. A mean pole position for these four sites has been calculated and referred to as the mean 'Pliocene mud' direction given in table 1.

The pliocene pole position from Baja California is distinctly anomalous as can be seen by a comparison of this pole position with other data from Upper Tertiary rocks of North America (fig. 3b and table 2). By contrast, the circle of confidence about the pole position for the Recent muds includes the rotation axis, the geocentric dipole and the pole determined from the local magnetic elements (fig. 3a).

Table 1
Paleomagnetic data – Baja California muds.

	No. of sites	No. of samples	D	I	K	α_{95}	Long.	Lat	δM	δP
Modern muds	16	16	19°	57°	25.2	7.5	50	79	11	8
Pliocene muds										
WRB A	24	30	53°	49°	9.2	10.3	306	22	14	9
WRB B	5	11	48°	38°	26.2	15.2	24	46	18	11
WRB C	5	6	33°	41°	17.0	19.1	18	60	17	19
Canon Rojo fines	8	14	54°	48°	17.1	13.8	36	44	18	12
Mean Pliocene muds		4	41°	43°	82.0	10.2	25	53	13	8

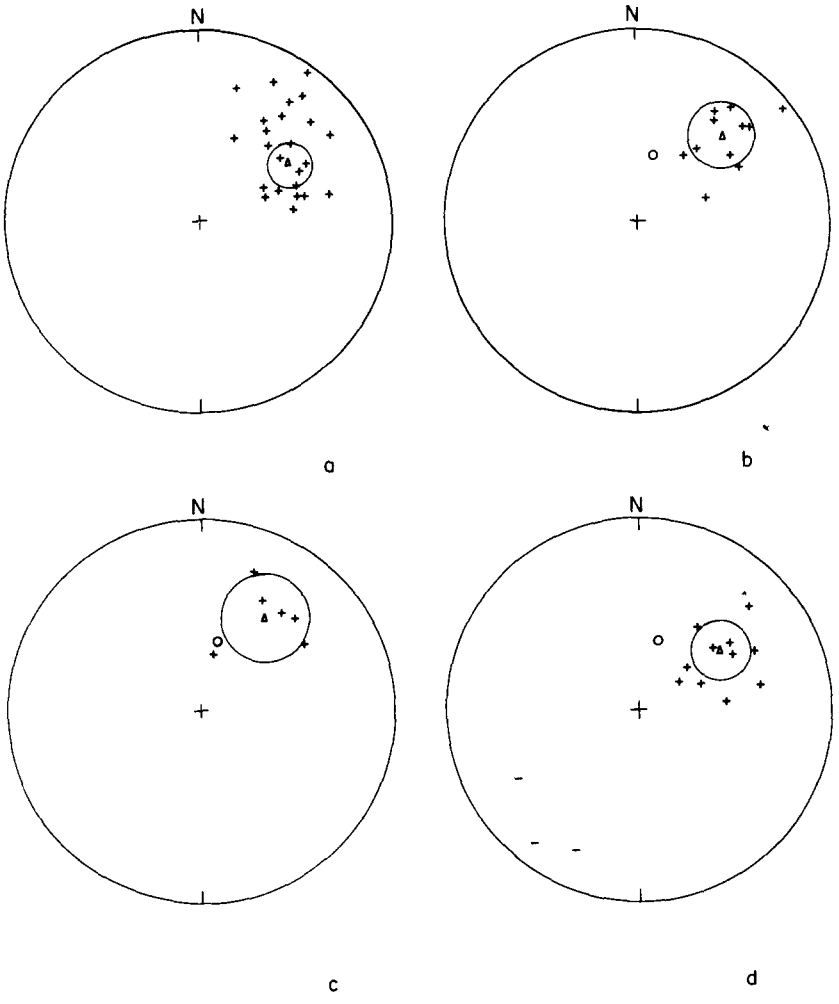


Fig. 2(a) Direction of magnetization in Pliocene mud (WRBA) after A.C. cleaning. (b) Direction of magnetization in Pliocene mud (WRBB) after A.C. cleaning. (c) Direction of magnetization in Pliocene mud (WRBC) after A.C. cleaning. (d) Direction of magnetization in fine-grained Pliocene sand (Canon Rojo) after A.C. cleaning.

As the statistical analyses of the direction obtained from the Recent samples (fig. 1, fig. 3a, table 1) indicate that the magnetization of these materials reflect the field at the time of deposition, we conclude that the Pliocene units are biased by post-depositional changes in the sediments, or the earth's field at the time of deposition was significantly different in direction from the present field, or there has been movement of the collection site with respect to the magnetic field since the magnetization was acquired. When the possible post-depositional changes such as compaction and chemical change are examined, it is found that these cannot adequately account for the deviation of the Pliocene magnetic direction from the present field or from the Tertiary field based on other North American data. Compaction tends to reduce the inclination and although some reduction in inclination between Recent and Pliocene units is observed the main cause of the divergence is not due to a difference in inclination but to a difference in the declination. Post-depositional changes do not explain the divergence because these would tend to bias the data toward the present field.

Another possible explanation for the anomalous

Table 2
Upper Tertiary pole positions from North America.

	Lat.	Long.	Reference [6]
Ellensburg formation (10.1 my)	85°	115° W	Irving 11.104 (Torreson et al., 1949
Columbia River basalts	85°	43° W	Irving 11.106 (Campbell and Runcorn, 1956)
Canadian basalts	85°	150° E	Irving 11.107 (du Bois, 1959)
Rio Grande flows, N. Mexico	80°	121° E	Ozima et al. (1967)
Oregon plateau basalts	80°	32° E	Watkins (1965)
Columbia River basalts	87°	171° W	Watkins (1965)
Baja California, Pliocene muds	53°	25° W	Present study

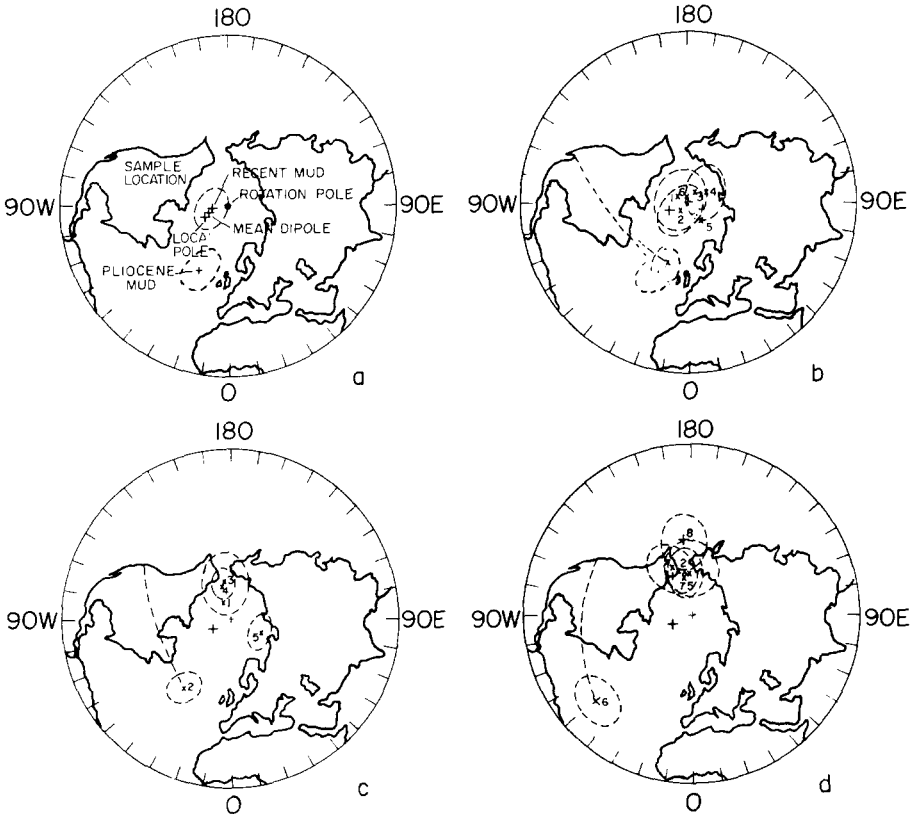


Fig. 3. (a) Mean pole position – Recent mud and Pliocene mud. Pole positions from North America (numbers as in tables 2–4), (b) Upper Tertiary, (c) Lower Tertiary, (d) Cretaceous (uncertainty limit for 3, 9 not available).

Pliocene direction is that the sampling represents a short time period in which there was a fluctuation of the magnetic field due to secular variation. This interpretation seems unlikely because of the number of localities samples (four) and because the thickness of the stratigraphic sequence (450 feet of fine-grained muds at WRBA) seems to represent a considerable amount of geologic time. Moreover, one group of samples (from a single site at Canon Rojo) includes a few magnetically reversed directions which further support the argument that the time span examined involves too long a period for the deviations to be due to secular variation alone. We, therefore, conclude that the mean position obtained is characteristic of the Pliocene for Baja California.

3. Possible interpretation

One possible explanation for the difference is that the Baja peninsula has undergone crustal movement of that of the rest of the North American continent. It has long been suggested that the Franciscan Formation and granitic material west of the San Andreas fault are slices of the continental mass which have been translated northwestward along the California coast. More recently, it has been suggested that the Gulf of California has been opened by rifting and that northwestward translation and clockwise rotation of at least the southern half of Baja California may have taken place [4]. Geometric considerations, however, limit the amount of rotation to a maximum of 12°. A northwestward translation of Baja California of about 12° along a fault system similar to that of the San Adreas has been suggested [4] but simple translation could not account for the observed pole position. The observed position is best explained by assuming that, since the rocks acquired their magnetization, the area has undergone a clockwise rotation of about 30°. As the rocks examined magnetically are Pliocene and the Gulf is believed to have been opened by Miocene time, at the latest [4], it is seen that both geometrical and time considerations suggest that the crustal movement postulated [4] could not have involved Baja California as a single unit.

In western Oregon, the Siletz River volcanics of Eocene age and in the San Francisco Bay area, the Franciscan rocks of Cretaceous age [5], have anomalous pole positions (see fig. 3c and 3d and tables 3 and 4). The divergence of these positions from the cluster of pole positions for other North American rocks of the appropriate ages is quite large. For the Franciscan rocks the divergence is about 110°. The divergence is in the same sense for each case, that is the pole positions seem to have been displaced primarily by a clockwise rotation at the collection site.

The paleomagnetic data from these three sites along the Pacific coast of North America are considered to be valid on the basis of the magnetic stability tests applied. Although the paleomagnetic data are not specific they do support the structural interpretation that parts of the Pacific coast have undergone significant crustal rotation independent of the North American continental mass. The directional differences are large and may indicate that parts of the west coast have undergone rotations in small independent blocks rather than in large, coherent units.

Certainly no simple structural movements can account for the paleomagnetic data. However, anomalous data from rocks of three different ages and from three different places suggests that differential rotations have taken place.

Table 3
Lower Tertiary pole positions – North America.

	Lat.	Long.	Reference [7]
Green River	78°	150° W	Irving 11.100 (Torreson et al., 1949)
Siletz River, Wash.	37°	49° W	Irving 11.098 (Cox, 1957)
Spanish Peaks, Colo.	75°	170° W	Larson and Strangway (1968)
Front Range Intrusive, Colo.	72°	166° W	McMahon and Strangway (1967)
S.E. Arizona volcanics (25–30 my)	75°	11° E	Strangway, in preparation

Table 4
Cretaceous pole positions from North America.

	Lat.	Long.	Reference
Montergian Hills, Que.	65N	157W	Irving 10:11 (Larochelle, 1961)
Mt. Megantic, Que.	69N	172E	Irving 10:12 (Larochelle, 1962)
Granite plutons, Sierra Nevada	69N	165W	Gromme and Merrill (1965)
Isachsen diabase	69N	180E	Larochelle and Black (1963)
Boulder batholith	76N	167E	Colville, 1961 (in Hanna, 1967)
Franciscan fm., Calif.	26N	50W	Gromme and Gluskoter (1965)
Elkhorn Mtns., Mont.	69N	171W	Hanna (1967)
Guadeloupe Mtn. and Bucks bath, Sierra Nevada	54N	174W	Gromme et al. (1967)
Mesaverde fm., Wyo. and Utah	70N	177E	Kilbourne (1967)

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References

[1] T.R. Walker, Formation of red beds in modern and ancient deserts, Geol. Soc. Am. Bull. 78 (1967) 353.

[2] D.W. Strandgway, T.R. Walker, E.E. Larson and B.E. McMahon, The magnetic properties of modern red sediments, Geol. Soc. Am. Bull., in press.

[3] E. Irving and A. Major, Post-depositional detrital remanent magnetization in a synthetic sediment, Sedimentology 3 (1964) 135.
R.F. King and A.I. Rees, Detrital magnetism in sediments: an examination of some theoretical models, J. Geophys. Res. 71 (1966) 561.

[4] W. Hamilton, Origin of the Gulf of California, Geol. Soc. Am. Bull. 72 (1961) 1307.
W. Hamilton and W.B. Meyers, Cenozoic tectonics in the Western United States, Rev. Geophys. 4 (1966) 509.
G.A. Rusnak, R.L. Fisher and F.P. Shepard, Bathymetry and faults of Gulf of California, A.A.P.G., Geol. Mem. 32 (1964) 144.
E.C. Allison, Geology of areas bordering Gulf of California. A.A.P.G., Geol. Mem. 32 (1964) 3.

[5] A. Cox, Remanent magnetism of lower to middle Eocene basalt flows from Oregon, Nature 179 (1951) 685.
C.S. Grommé and H.J. Gluskoter, Remanent magnetization of spilite and diabase in the Franciscan formation, Western Marin County, California, J. Geol. 73 (1965) 74.

[6] E. Irving, Paleomagnetism (Wiley, New York, 1964) 399 pp.
M. Ozima, M. Kono, I. Kaneoka, H. Kinoshita, K. Kobayashi, T. Nagata, E.E. Larson and D. Strangway, Paleomagnetism and potassium-argon ages of some volcanic rocks from the Rio Grande gorge, New Mexico, J. Geophys. Res. 72 (1967) 2615.
N.D. Watkins, Paleomagnetism of the Columbia Plateaus, J. Geophys. Res. 70 (1965) 1379.

[7] E. Irving, Paleomagnetism (Wiley, New York, 1964) 399 pp.
B.E. McMahon and D.W. Strangway, Investigation of Kiaman magnetic division in Colorado Redbeds, Geophys. J. 15 (1968) 265.
E.E. Larson and D.W. Strangway, Magnetization of the Spanish Peaks Dike Swarm, Colorado and Shiprock Dike, New Mexico, J. Geophys. Res. 74 (1969) 1505.

[8] E. Irving, Paleomagnetism (Wiley, New York, 1964) 399 pp.
C.S. Grommé and R.T. Merrill, Paleomagnetism of Late Cretaceous granitic plutons in the Sierra Nevada, California; Further results. J. Geophys. Res. 70 (1965) 3407.
A. Larochelle and R.F. Black, An application of paleomagnetism in estimating the age of rocks, Nature 198 (1963) 1260.
C.S. Grommé and H.J. Gluskoter, Remanent magnetization of splite and diabase in the Franciscan formation, western Marin County, California, J. Geol. 73 (1965) 75.
W.F. Hanna, Paleomagnetism of upper Cretaceous volcanic rocks of southwestern Montana, J. Geophys. Res. 72 (1967) 595.
D.E. Kilbourne, Paleomagnetic properties of late Cretaceous sediments, Trans, Am. Geophys. Union 48 (1967) abstract.
C.S. Gromme, R.T. Merrill and J. Verhoogen, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California and its significance for Polar Wandering and continental drift, J. Geophys. Res. 72 (1967) 5661.